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$$\begin{aligned}& \int \frac{5x+2}{x-3} dx \\& 5 \int \frac{x+\frac{2}{5}}{x-3} dx \\& 5 \int \frac{x-3}{x-3} dx + \int \frac{\frac{17}{5}}{x-3} dx \\& 5 \int dx + \frac{17}{5} \int \frac{dx}{x-3}\end{aligned}$$

$$\text{Stel } t = x - 3 \Rightarrow dt = dx$$

$$\begin{aligned}&= 5 \int dx + \frac{17}{5} \int \frac{dt}{t} \\&= 5x + \frac{17}{5} \ln |t| + c \\&= 5x + \frac{17}{5} \ln |x-3| + c\end{aligned}$$

References